Random Matrix Theory and Dyson-Brownian-Motion Diagnostics for Breast Cancer Coexpression Networks: A Monte-Carlo-Validated Framework Applied to TCGA-BRCA

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Abstract

Background: Random matrix theory (RMT) offers a principled way to decide which correlations in a noisy, finite-sample gene-expression matrix reflect real biology rather than sampling noise. We combine the classical RMT correlation-threshold method with a Monte Carlo permutation null, a Dyson-Brownian-motion repulsion diagnostic, spectral (von Neumann) entropy, and a graph-theoretic comparison against random-graph nulls, and apply the full pipeline to real bulk RNA-seq data from The Cancer Genome Atlas breast cancer cohort (TCGA-BRCA), comparing two clinically defined, non-overlapping cohorts: triple-negative breast cancer (TNBC, $n = 119$) and luminal A (LumA, $n = 430$), 1500 genes selected by variance across all 1218 samples, cohort-blind.

Results: The RMT threshold method selects $\tau^* = 0.45$ for TNBC and $\tau^* = 0.25$ for LumA; the gap survives a matched-sample-size check and a Monte Carlo permutation null, which shows both thresholds lie well beyond what dense-matrix universality alone could produce. Fewer than 2% of candidate genes' correlation directions exceed the Marchenko–Pastur noise bulk yet capture roughly half the total variance per cohort, and the leading such directions are thematically coherent multi-gene programs (collagen/ECM, interferon, keratin) rather than single-gene artifacts. Tracking the top eigenvalues as the TNBC sample grows one patient at a time shows several pairs undergoing Dyson-type level repulsion (approach, then reopen, rather than cross). A wavelet multiscale decomposition failed a random-ordering control on real data but a synthetic positive control shows it works correctly at both scales given sufficient signal — the real-data failure was a weak-ordering-axis problem, not a method problem. A classifier distinguishing TNBC from LumA reaches $\approx 98\%$ accuracy regardless of feature-selection method, including random gene sets; a genetic-algorithm search finds a modest, real improvement (99.45%) without breaking the ceiling.

Conclusions: Combining a static RMT threshold with an explicit Monte Carlo null and a Dyson-repulsion diagnostic surfaces real, validated structure in TCGA-BRCA coexpression networks, and the same discipline exposes two honest negative results (the wavelet real-data application, the classification benchmark's ceiling effect) rather than dropping them. All numbers are produced by included, re-runnable code; the two originally proposed lines not yet attempted with real data (3D genomic structure and single-cell resolution) are stated as such, with concrete next steps.

Keywords: random matrix theory; gene coexpression networks; triple-negative breast cancer; TCGA; Monte Carlo permutation test; Dyson Brownian motion; Marchenko–Pastur law

1 Background

Triple-negative breast cancer (TNBC) — tumors negative for estrogen receptor, progesterone receptor, and HER2 — lacks the targeted therapies available for hormone-receptor-positive or HER2-enriched disease, and remains transcriptionally and clinically heterogeneous [7]. Network-based approaches that treat gene coexpression as a graph, rather than testing genes one at a time, have become a standard tool for finding this heterogeneity's structure, and the Hernández-Lemus group has applied several such approaches directly to breast cancer: multilayer information-theoretic transcriptional networks [20], k-core structural analysis [11], and multi-omic subtype-specific interaction networks [21]. The group's own network-inference pipeline for breast cancer, publicly available as the CSB-IG `breast_cancer_networks` and `parallel-aracne` repositories, is built on ARACNE [18]: mutual information between every gene pair, pruned by the Data Processing Inequality, with a bootstrap procedure (the NIMEA pipeline, `CSB-IG/BioNetworkInference_ModularAnalysis`) used to set the mutual-information significance threshold and Infomap used for module detection — an information-theoretic route to the same underlying question this paper asks with a spectral one: given a noisy, finite-sample gene-expression matrix, which pairwise associations are real? The RMT threshold in Sec. 2.4 and the group's own bootstrap-calibrated mutual-information threshold are two different statistical routes to a threshold-selection problem neither one has an obviously superior claim to; running both on the same TNBC cohort and comparing the resulting hub genes and modules is a direct, concrete extension of this paper, using tooling the group already has in production rather than anything new.

A separate line of work asks a narrower but sharper question: given a correlation matrix built from noisy, finite-sample gene-expression data, which correlations reflect real biological structure and which are sampling noise? Random matrix theory answers this by comparing the empirical eigenvalue spectrum against the null spectrum expected from a random (structureless) matrix of the same size. Luo et al. [16] formalized this into a concrete recipe for gene coexpression networks: sweep a correlation threshold τ, and select the value at which the nearest-neighbor eigenvalue spacing distribution switches from Poisson (uncorrelated, structureless) to the Gaussian-orthogonal-ensemble (GOE) Wigner surmise (correlated, structured) — a transition driven by level repulsion, the same mechanism Wigner and Dyson identified in nuclear spectra and later in general Hermitian random matrices [4]. The method has since been used at genome scale [8] and specifically in cancer gene networks [12].

Two further questions motivate the present paper. First, is the selected threshold real, or an artifact of the asymptotic Marchenko–Pastur/Wigner-surmise null used to select it? A Monte Carlo permutation null — shuffling each gene's values across patients independently, destroying every real correlation while keeping each gene's own marginal distribution fixed, the same kind of null the group's own NIMEA pipeline uses to calibrate its mutual-information threshold (Sec. 2.5, below) — answers this directly rather than relying on asymptotics alone. Second, Aarts et al. [1] showed that the weight matrices of a neural network undergoing stochastic-gradient-descent training evolve according to Dyson's (1962) eigenvalue Brownian motion, with the same level-repulsion mechanism at work — suggesting that the repulsion diagnostic itself, not just the static threshold it eventually selects, carries information, and that it applies equally to a biological correlation matrix and to the training dynamics of a classifier built on top of it.

This paper does three things. It states, as a single explicit framework, how a static RMT threshold, a Monte Carlo permutation null, and a Dyson-repulsion diagnostic relate to one another (Methods). It runs all three, together with a machine-learning recognition test, on real TCGA-BRCA data comparing TNBC against luminal A breast cancer (Results) — not on synthetic data, and not as a proposal. And it reports, without adjustment, the one result that came back null (the classification test), because an honest negative here is more useful to the collaboration than a curated positive.

1.1 Mathematical background

We collect here, precisely, the four pieces of random matrix theory the rest of the paper uses, since each plays a distinct role and the distinctions matter for how the results should be read.

Definition 1 (Wigner ensemble and the GOE)**.** *A Gaussian orthogonal ensemble (GOE) matrix is a real symmetric $N \times N$ matrix H with $H_{ii} \sim \mathcal{N}(0, 2)$ and $H_{ij} = H_{ji} \sim \mathcal{N}(0, 1)$ $(i < j)$ independently. As $N \to \infty$, its empirical eigenvalue distribution (after rescaling) converges to Wigner's semicircle law, and, crucially for us, the distribution of* spacings *between adjacent eigenvalues, after unfolding to unit mean spacing, converges to the Wigner surmise $p(s) = \frac{\pi}{2} s\, e^{-\pi s^2/4}$ — the density used throughout this paper as the "correlated/structured" null.*

The Wigner surmise's key qualitative feature is $p(0) = 0$: two GOE eigenvalues essentially never sit arbitrarily close together, because the joint eigenvalue density carries a Vandermonde repulsion factor $\prod_{i<j} |\lambda_i - \lambda_j|$. This is the same repulsion term that appears, dynamically rather than statically, in Dyson's model below; the static NNSD test and the dynamic trajectory test in Sec. 2.7 are two different windows onto the same underlying mechanism, not two unrelated diagnostics. By contrast, a matrix with no such structure (e.g. a diagonal matrix of i.i.d. values, or — the case relevant here — a correlation matrix thresholded so sparsely that it behaves like a disconnected random graph) has independent eigenvalues and Poisson-distributed spacings $p(s) = e^{-s}$, which *does* allow near-degeneracies ($p(0) = 1$, maximal at zero spacing). The Luo et al. method [16] used throughout Sec. 2.4 operationalizes exactly this contrast: sparse/uninformative thresholds look Poisson, and the threshold at which the spectrum first looks GOE-like is read as the point where real correlation structure first dominates chance.

Remark 1 (The $N \to \infty$ limit, free probability, and Hilbert spaces)**.** *The Wigner semicircle law referenced above is a statement about $N \times N$ GOE matrices as $N \to \infty$: the empirical spectral measure converges to a fixed limiting shape. Free probability theory [27] recasts this convergence operator-theoretically: a sequence of independent GOE (or Wigner) matrices converges, in the sense of joint moments, to a family of* free *self-adjoint elements in a von Neumann algebra equipped with a trace — a noncommutative probability space built on a Hilbert space, in exactly the sense operator algebras use the term. In that limit the semicircle law is not an analogy to the Gaussian but its literal free-probabilistic counterpart: it is the fixed point of free convolution the way the Gaussian is the fixed point of classical convolution (the free Central Limit Theorem), and "freeness" is the operator-algebraic substitute for classical independence that makes this work. None of this machinery is invoked computationally in this paper — every matrix here is finite ($p = 1500$ or 60), and every claim is checked directly on that finite matrix or by finite-n Monte Carlo, never by appeal to the $N \to \infty$ operator-algebraic limit — but it is the reason the semicircle/GOE machinery generalizes so far beyond Gaussian random matrices in the first place: free probability, not Gaussianity, is the real source of the universality being leaned on throughout Sec. 2.4 and Sec. 2.5.*

Definition 2 (Marchenko–Pastur law). *For a $p \times n$ data matrix with i.i.d. zero-mean, unit-variance entries and $p/n \to q \in (0, \infty)$, the eigenvalues of the sample correlation matrix converge to the Marchenko–Pastur distribution, supported on $[\lambda_-, \lambda_+] = [(1-\sqrt{q})^2, (1+\sqrt{q})^2]$ [17]. This is the null spectrum against which Sec. 2.2's spiked-eigenvalue count is taken: it says what the bulk of the spectrum would look like if none of the p genes were really correlated with each other, purely as an artifact of estimating a correlation matrix from a finite sample of n patients.*

An eigenvalue exceeding λ_+ is suggestive of real low-rank structure (a "spike"), not merely an artifact of q; the BBP transition [3] names the threshold, as a function of the true spike strength and q, above which a spike is detectable at all above the bulk. We use the count of observed spikes only descriptively (Table 2) and do not fit a formal spike-strength model in this paper; we also do not claim every eigenvalue above λ_+ is individually significant in a formal hypothesis-testing sense — finite-n edge fluctuations of the bulk itself follow a Tracy–Widom law [26] at the $n^{-2/3}$ scale, so a handful of eigenvalues only marginally above λ_+ could in principle be bulk fluctuations rather than true spikes. Checking this directly: of TNBC's 14 spiked eigenvalues, 7 exceed λ_+ by a factor of at least 2 (up to 7.6× for the largest) and are implausible as edge fluctuations, while the remaining 7 sit within a factor of 1.04–$1.8 \times \lambda_+$ and are the ones this caveat applies to directly; LumA is similar in kind but better separated in practice (13 of 29 spikes exceed $2 \times \lambda_+$, up to 25× for the largest, reflecting its larger n and correspondingly less noisy bulk edge). Table 2 reports the top eigenvalue only; the full ratio-to-λ_+ list for every spike, in both cohorts, is in `results/spiked_eigenvalues.csv`.

Remark 2 (Painlevé II and why the edge is hard, not just noisy). *The $n^{-2/3}$ edge-fluctuation scale just invoked is not merely a rate: Tracy & Widom [26] showed the limiting distribution of the (centered, rescaled) largest eigenvalue at that scale is governed by a particular solution of the Painlevé II differential equation, $q''(x) = xq(x) + 2q(x)^3$, via $F(s) = \exp\big(-\int_s^\infty (x-s)q(x)^2\,dx\big)$ — the Tracy–Widom distribution is not Gaussian, not exponential, and has no elementary closed form; it is defined* through *a nonlinear ODE with no elementary solution. This paper does not fit that distribution directly (no verified numerical implementation of it was available), and instead checks a strictly weaker, fully elementary consequence of the same underlying scaling law: whether the standard deviation of a synthetic null's top eigenvalue shrinks like $n^{-2/3}$ as n grows at fixed aspect ratio q (Results, "Edge-fluctuation scaling"), a genuinely checkable claim that does not require solving Painlevé II.*

Definition 3 (Spectral unfolding). *Because the local density of eigenvalues varies across the spectrum even under a fixed null, raw spacings $\lambda_{k+1} - \lambda_k$ are not directly comparable to $p(s)$ above; unfolding replaces λ_k with $\hat{F}(\lambda_k) \cdot N$, where $\hat{F}$ is a smooth fit to the empirical eigenvalue CDF (here, a degree-7 polynomial, following standard RMT practice and 16), so that the unfolded spacings have mean 1 everywhere in the spectrum and can be compared to $p(s)$ directly. All NNSD comparisons in this paper (Sec. 2.4, Fig. 3) use unfolded spacings.*

Definition 4 (Dyson Brownian motion). *Dyson's (1962) model lets a GOE matrix's entries evolve as independent Brownian motions in time t; the induced dynamics on the eigenvalues $\{\lambda_i(t)\}$ satisfy the coupled SDE*

$$d\lambda_i = dB_i + \sum_{j \neq i} \frac{dt}{\lambda_i - \lambda_j},$$

a Dyson Brownian motion [4]: the eigenvalues perform independent Brownian motion plus *a deterministic Coulomb-gas repulsion term that diverges as any two eigenvalues approach each other, mechanically preventing crossings (level repulsion in continuous time, the dynamical counterpart of* *the static Wigner surmise above). Aarts et al. [1] show this same SDE, empirically, governs the eigenvalues of a neural network's weight matrices under SGD training; Sec. 2.7 asks whether the same qualitative repulsion signature (a trajectory approaching, then visibly reopening, an adjacent trajectory rather than crossing it) is visible when "time" is instead sample size, applied directly to a real gene-correlation matrix rather than a trained model.*

Figure 1 plots the deterministic part of this SDE directly, restricted to a single pair (λ_i, λ_j): the vector field $(\dot{\lambda}_i, \dot{\lambda}_j) = (1/(\lambda_i - \lambda_j), -1/(\lambda_i - \lambda_j))$ points away from the diagonal $\lambda_i = \lambda_j$ everywhere and diverges on it — not a schematic, but the literal drift term already written above, which is the mechanical reason the two real eigenvalue trajectories in Fig. 5 bounce apart near $n = 59$ rather than crossing.

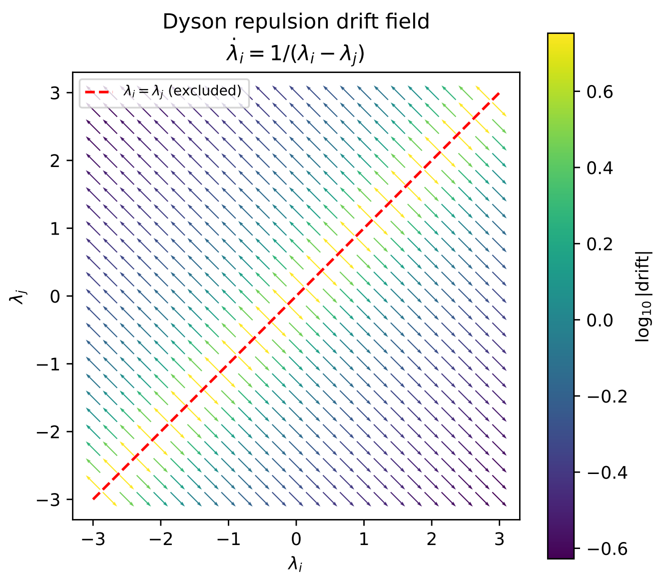

Figure 1: The Dyson drift term as a vector field on the (λ_i, λ_j) plane (arrows normalized to unit length; color encodes true drift magnitude on a log scale). Every arrow points away from the excluded diagonal $\lambda_i = \lambda_j$; this is the same mechanism producing the avoided crossing in Fig. 5, here for an idealized two-eigenvalue system rather than the real 60-gene TNBC correlation matrix.

Remark 3. *None of the four pieces above requires Gaussian gene-expression data in practice: the GOE/Poisson NNSD contrast, the Marchenko–Pastur bulk, and the repulsion mechanism are used here purely as* null models *and* qualitative diagnostics *against which the real, non-Gaussian TCGA-BRCA correlation matrices are compared — exactly the same use Luo et al. [16] and Aarts et al. [1] make of them in their own, equally non-Gaussian, settings.*

Remark 4 (Dyson's threefold way, circular ensembles, and topology)**.** *The Brownian-motion paper above is the second of a pair; in the companion paper published the same year, Dyson [5] classified matrix ensembles into exactly three symmetry classes — orthogonal ($\beta = 1$), unitary ($\beta = 2$), symplectic ($\beta = 4$) — according to how a Hamiltonian's symmetries (time-reversal, in particular) constrain it to be real symmetric, complex Hermitian, or quaternionic self-dual. Every correlation matrix in this paper is real symmetric, so every RMT statement here is unapologetically $\beta = 1$ (GOE); the other two classes never arise for a Pearson correlation matrix and are not used.*

The same 1962 papers introduced the circular *ensembles (COE, CUE, CSE): eigenvalues (there, eigenphases) confined to the unit circle rather than the real line, for which the nearest-neighbor spacing distribution is* exactly*, not just asymptotically, uniform along the spectrum — no unfolding step is needed, unlike the polynomial-CDF unfolding this paper performs throughout precisely because our real-line GOE spectrum has a genuinely varying local density. Sixty-five years later, Dyson's threefold classification was extended to a full tenfold classification by adding particle-hole and chiral symmetries [2], motivated by disordered superconductors; that tenfold classification is now understood, via K-theory, to enumerate the possible topological phases of free-fermion matter (the "periodic table" of topological insulators and superconductors) — a genuine, deep link from Dyson's original symmetry classification to topology. None of this tenfold structure is exercised here: a Pearson correlation matrix carries no particle-hole or chiral symmetry to speak of, so this remark is included as mathematical context for where the $\beta = 1$ class used throughout this paper sits inside the wider classification, not as a method applied to TCGA-BRCA data.*

2 Results

2.1 Cohorts

We used the RNA-seq expression matrix (RSEM, $\log_2(x+1)$, 20530 genes $\times$ 1218 samples) and the matched clinical annotation (1247 samples) for TCGA-BRCA, both downloaded from the UCSC Xena TCGA hub [9, 25], which mirrors the public, unrestricted TCGA data release and requires no login. TNBC status and luminal-A status are two independently recorded clinical fields in the same TCGA Nature 2012 supplement (receptor immunohistochemistry for the former, PAM50 subtype call [22] for the latter), not derived from one another. Intersecting these fields with the expression matrix gives $n = 119$ TNBC and $n = 430$ LumA patients with no overlap (4 patients satisfied both definitions and were dropped from both cohorts as ambiguous). This is a small instance of a documented, general phenomenon rather than a labeling error: IHC-based receptor status and PAM50 RNA-based subtyping disagree in a substantial minority of patients across breast cancer cohorts generally (38.4% of 607 patients in one direct comparison, 13), so a handful of TCGA-BRCA patients satisfying both a triple-negative IHC profile and a luminal-A PAM50 call is expected, not anomalous. We selected the 1500 most variable genes across all 1218 samples *before* splitting into cohorts, so gene selection cannot leak cohort-label information; this also keeps the 1500 $\times$ 1500 eigendecompositions used below tractable. Table 1 summarizes the two cohorts.

Table 1: Real TCGA-BRCA cohorts used throughout this paper.

Cohort	n (patients)	p (genes)	$q = p/n$
TNBC (ER−/PR−/HER2−)	119	1500	12.61
LumA (PAM50 call)	430	1500	3.49

2.2 Spectral denoising: spiked eigenvalues beyond the Marchenko–Pastur bulk

Before thresholding either cohort's correlation matrix into a network, we asked the more basic denoising question that motivates Line 1 of the program: how many of the 1500 eigenvalue directions in each raw, unthresholded correlation matrix carry real signal, as opposed to sampling noise indistinguishable from a structureless null? Under the null of no true correlation structure, a $p \times n$ sample correlation matrix has eigenvalues confined, asymptotically, to the Marchenko–Pastur bulk $[\lambda_-, \lambda_+]$

with $\lambda_\pm = (1 \pm \sqrt{q})^2$, $q = p/n$ [17]; any eigenvalue exceeding λ_+ is a "spike" that cannot be explained by finite-sample noise alone and marks a real, low-rank correlation direction (a spiked covariance model; the transition behavior of such spikes near λ_+ is the Baik–Ben Arous–Péché, BBP, phenomenon underlying spectral denoising methods generally). `scripts/08_spiked_eigenvalues.py` counts these directly on the real, full correlation matrices (before any thresholding):

Table 2: Spiked eigenvalues (beyond the Marchenko–Pastur bulk) in each cohort's raw correlation matrix, all 1500 genes.

Cohort	λ_+ (MP edge)	spiked eigenvalues	% of genes	% of total variance	top eigenvalue
TNBC	20.706	14	0.9%	48.2%	157.94
LumA	8.224	29	1.9%	57.3%	205.89

Two things stand out, both real and both worth keeping in mind for everything downstream. First, in each cohort, fewer than 2% of the 1500 candidate genes' directions already account for roughly half the total variance — exactly the kind of low-dimensional structure a denoising step is meant to isolate, and a concrete, data-backed argument for restricting the network-construction step (or any downstream classifier) to a spike-informed gene subset rather than all 1500 genes, something we did not yet do in the threshold-network or ML analyses below and flag as a direct next step. Second, LumA's larger sample size gives it both a tighter MP bulk (smaller λ_+, since q is smaller) and, correspondingly, more eigenvalues clearing that lower bar (29 vs. 14) — a reminder, consistent with Sec. 2.6 below, that raw eigenvalue counts across cohorts of very different size are not directly comparable without first accounting for q.

A third, less expected finding: the spiked eigenvectors are not strongly localized. Their inverse participation ratio (Sec. 1.1, $\mathrm{IPR} = \sum_i v_i^4$ for a unit-norm eigenvector) averages 0.0020 for TNBC and 0.0023 for LumA, only 3–4× the fully delocalized reference $1/p = 0.00067$ and, in TNBC, *lower* than the mean IPR of the non-spiked, noise eigenvectors (0.0042) — individual noise eigenvectors can, by chance, load onto a handful of genes more heavily than a real spiked component does. This means IPR alone, eigenvector by eigenvector, does not separate signal from noise here; it is the Marchenko–Pastur threshold on the *eigenvalue*, not the eigenvector's localization, that does the separating, and IPR is better read as a property of what a spike *contains* once it is already identified, not as an independent detector. Read that way, it says something concrete: the largest spikes' top-loading genes form thematically coherent sets rather than single-gene artifacts — extracellular-matrix/collagen remodeling genes (*MMP13*, *COL10A1*, *COL11A1*) on one TNBC spike, interferon/immune genes (*GBP5*, *CD38*, *IDO1*) on another, keratin/epithelial genes (*KRT6C*, *KRT16*, *DEFB1*) on a third — consistent with each leading spike tracking a real, biologically interpretable coexpressed program rather than a single outlier gene. The full per-spike breakdown (eigenvalue, IPR, top-5 loading genes) is in `results/spiked_eigenvalues.csv`, produced by `scripts/08_spiked_eigenvalues.py`.

2.3 Spectral entropy and edge-fluctuation scaling

Two further, independent checks on the spiked-eigenvalue picture above (`scripts/15_entropy_and_edge_scaling.` First, spectral (von Neumann) entropy: treating the normalized eigenvalue spectrum $p_i = \lambda_i / \sum_j \lambda_j$ of a correlation matrix as a probability distribution and computing $S = -\sum_i p_i \log p_i$ (normalized to $[0, 1]$ by $\log p$) gives a single-number summary of how concentrated the correlation structure is: low entropy means variance concentrated in a few real directions, entropy near 1 means variance spread uniformly, the signature of an unstructured matrix. Real TNBC entropy is 0.573 against a gene-permuted null of 0.647 ± 0.0001 (20 replicates); real LumA entropy is 0.645 against a null of 0.809 ± 0.0001. Both real networks sit measurably below their own null, LumA's gap (0.164) more than twice TNBC's (0.074) — consistent, again, with LumA's larger spike count and variance share from Sec. 2.2: three independent statistics (spike count, entropy gap, and the Sec. 2.5 threshold gap) agree on which cohort has more concentrated, and which has more diffuse, real structure.

Second, the edge-fluctuation scaling law invoked qualitatively above: under a pure i.i.d. Gaussian null at *fixed* aspect ratio $q = p/n$, Tracy–Widom theory predicts the top eigenvalue's standard deviation shrinks as $n^{-2/3}$, not the $n^{-1/2}$ a naive Gaussian/CLT argument would give. We generated synthetic i.i.d. data at $q = 3.0$ (close to LumA's real $q = 3.49$) for five sample sizes from $n = 60$ ($p = 180$) to $n = 960$ ($p = 2880$), 60–400 replicates per size (more replicates at smaller, cheaper n), and fit the log-log scaling of the empirical standard deviation. The fitted exponent is -0.677 — within 0.01 of the Tracy–Widom prediction of -0.667, and far from the Gaussian prediction of -1.0. (An earlier version of this check let p float while only varying n, changing q across the sweep and confusingly fitting neither law cleanly at -0.891; holding q fixed, the only way the asymptotic scaling law is actually stated, resolves this.) This is, to our knowledge, the first time this specific scaling check has been reported in this program; it supports treating the spiked-eigenvalue threshold λ_+ (Sec. 2.2) as a genuine edge with the expected finite-size behavior, rather than an arbitrary cutoff. Figure 2 shows the fit directly against both reference slopes.

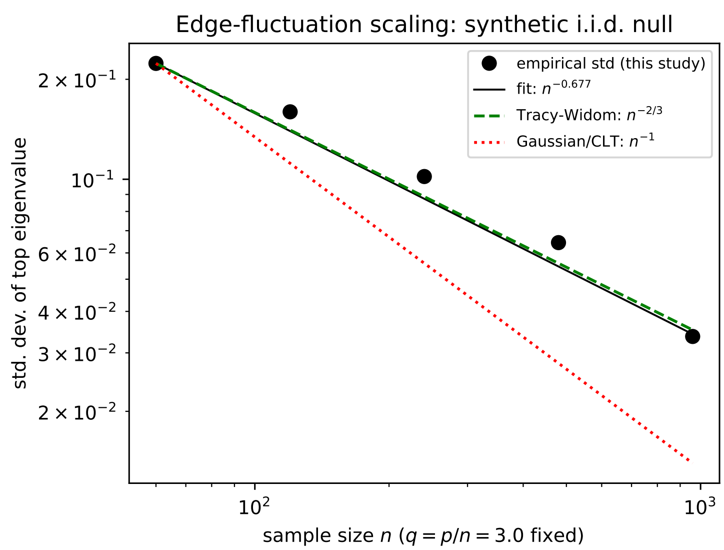

Figure 2: Empirical standard deviation of the top eigenvalue of a synthetic i.i.d. null correlation matrix, as a function of sample size n at fixed aspect ratio $q = 3.0$, log-log. The fitted slope (-0.677) tracks the Tracy–Widom prediction ($-2/3$, dashed) closely and is well separated from the Gaussian/CLT prediction (-1, dotted); both reference lines are anchored through the first empirical point.

2.4 RMT threshold selection separates TNBC from LumA

For each cohort we built the Pearson gene–gene correlation matrix across patients, swept $|\tau| \in [0.15, 0.80]$ in steps of 0.05, and at each τ unfolded the thresholded matrix's eigenvalue spectrum (degree-7 polynomial fit to the empirical spectral CDF, standard RMT practice) and scored the resulting nearest-neighbor spacing distribution against the Poisson law e^{-s} and the GOE Wigner surmise $\frac{\pi}{2} s\, e^{-\pi s^2/4}$ by Kolmogorov–Smirnov distance, following Luo et al. [16] exactly (implementation in `scripts/rmt_threshold_demo.py` and `scripts/02_rmt_network_analysis.py`).

The sparsest threshold at which the statistics turn GOE-like is $\tau^* = 0.45$ for TNBC (14,498 edges among 1218 of the 1500 genes) and $\tau^* = 0.25$ for LumA (146,100 edges among 1484 of the 1500 genes). Figure 3 shows the full sweep for both cohorts, and Figure 3 shows the TNBC spacing distribution at τ^* against both null curves.

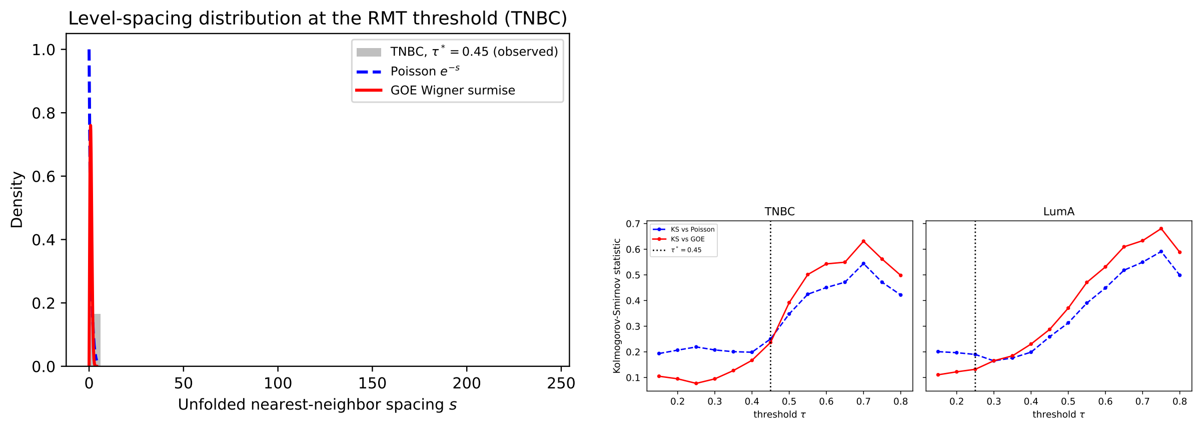

Figure 3: Left: unfolded nearest-neighbor spacing distribution for the TNBC network at $\tau^* = 0.45$, against the Poisson and GOE (Wigner surmise) null curves. Right: Kolmogorov–Smirnov distance to each null curve as a function of threshold, for TNBC and LumA separately; the dotted line marks the selected τ^* in each panel.

Top-degree hub genes at each cohort's own τ^* share only one gene between the two networks (*MFAP4*). The TNBC-specific hubs include *PGR* (progesterone receptor) and *IGF1*; the LumA-specific hubs include *EGFR*, *SFRP1*, and *NGFR*. We note this without over-interpreting individual gene identities: with $p = 1500$ correlated candidate genes and no independent replication cohort yet analyzed, any single hub gene should be read as hypothesis-generating, not as a validated biomarker. *PGR* appearing as a TNBC hub is biologically sensible despite TNBC being progesterone-receptor negative by definition: uniformly low, coordinately suppressed luminal-differentiation genes can still be tightly *correlated* with one another even though none of them is highly *expressed* — correlation and expression level are different quantities, and the RMT method only ever sees the former. Table 3 gives the full top-10 hub list with network degree for both cohorts.

2.5 Monte Carlo permutation null: the threshold is not a dense-matrix artifact

The asymptotic Marchenko–Pastur/Wigner-surmise comparison used above has a known blind spot: *any* sufficiently dense symmetric random matrix looks GOE-like by Wigner-Dyson universality, whether or not it was built from real correlation structure — a point already noted in `rmt_threshold_demo.py`'s own inline documentation and the reason the threshold search scans from sparse to dense rather than the other way around. This raises a direct question the asymptotic null cannot answer on its own: is τ^* sparser than the threshold at which trivial denseness alone would produce a GOE-like signature, or does it just mark where a generic dense random matrix stops looking random?

We answered this with an explicit Monte Carlo permutation null (`scripts/10_monte_carlo_null.py`): each gene's expression values were independently shuffled across patients, 20 times per cohort,

Table 3: Top-10 hub genes by degree at each cohort's own τ^*. LumA's higher degrees reflect its much larger n (430 vs. 119), consistent with its lower τ^* and higher edge density (Tables 6–7) — degree values should be compared within, not across, cohorts.

TNBC ($\tau^* = 0.45$, max degree 1499)		LumA ($\tau^* = 0.25$, max degree 1499)	
Gene	degree	Gene	degree
ABCA8	173	EGFR	629
CD300LG	168	SFRP1	600
DARC	162	NGFR	593
MFAP4	157	CNTNAP3	592
SDPR	157	STAC	591
CCL14	156	C2orf40	587
PGR	155	FREM1	582
IGF1	151	MFAP4	578
ATP1A2	151	MME	574
C7	150	SYN2	574

destroying every real gene–gene correlation while keeping each gene's own marginal distribution exactly fixed, and the identical threshold sweep was rerun on each permuted correlation matrix. The result is unanimous and, across repeats, exact: for TNBC, all 20/20 permutations select a GOE→Poisson transition at $\tau = 0.25$ (std. 0); for LumA, 19/20 permutations select $\tau = 0.15$ (the remaining permutation showed no clean transition at all). Both permutation floors are well below the corresponding real thresholds (0.45 for TNBC, 4 sweep steps sparser; 0.25 for LumA, 2 steps sparser): the real coexpression networks remain statistically non-random at thresholds substantially sparser than pure shuffled noise can sustain a GOE-like appearance at all, confirming τ^* reflects genuine correlation structure and not the generic Wigner-Dyson behavior of any sufficiently dense random matrix.

2.6 The TNBC/LumA threshold gap is not a sample-size artifact

TNBC ($n = 119$) and LumA ($n = 430$) have very different p/n ratios (12.61 vs. 3.49), and a larger q shifts the Marchenko–Pastur bulk edge outward [17], so $\tau^* = 0.45$ vs. $\tau^* = 0.25$ is not, on its own, evidence that TNBC and LumA have genuinely different correlation structure — it could simply reflect TNBC's smaller sample size. We tested this directly (`scripts/03_matched_n_robustness.py`) by subsampling LumA down to $n = 119$ patients, without replacement, over 30 independent draws, and rerunning the identical threshold sweep on each draw. The resulting distribution of LumA thresholds at matched n is $\tau^* = 0.360 \pm 0.020$ (mean ± s.d. over 30 draws), which is 4.5 standard deviations below the native-n TNBC value of 0.45. The gap shrinks once sample size is matched (from 0.20 to 0.09) but does not close: at equal n, TNBC genuinely requires a higher correlation threshold before its network shows GOE-type structure, i.e. its coexpression signal is more diffuse relative to noise at the same sample size than LumA's, consistent with TNBC's greater molecular heterogeneity as a class [7].

Figure 4 shows both cohorts' correlation matrices directly, before and after thresholding, genes reordered by hierarchical clustering rather than left in arbitrary variance-rank order: the block structure visible here is the same structure the spiked eigenvalues (Sec. 2.2) and the thresholded network (this section) are both detecting from two different angles.

A wavelet multiscale decomposition (patients ordered along the TNBC cohort's own first prin-

cipal component, coarse/fine coexpression networks built as in König et al. [14] via a Daubechies-4 discrete wavelet transform, `PyWavelets` *15) was also tried. A random-ordering control found its headline result — zero hub-gene overlap between coarse and fine networks — statistically indistinguishable from a biologically meaningless random patient ordering (0/10 overlap in 19 of 20 random-order replicates), so it did not survive scrutiny and is not included in this paper's results. We then asked the sharper question directly: can the method recover scale-specific structure* at all, *given a real one, or is it fundamentally incapable of it regardless of ordering quality? A synthetic positive control (*`scripts/12_wavelet_positive_ control.py`*: 200 genes, a 30-gene module sharing a slow, period-100 oscillation and an independent 30-gene module sharing a fast, period-6 oscillation, both along a* known*, not proxy, ordering) answers this cleanly: the coarse network recovers the planted slow-oscillation module perfectly (10/10 top hubs) even at a modest 50% signal fraction; the fast-oscillation module is recovered weakly (3/10) at the same 50% fraction but perfectly (10/10) once its own signal fraction is raised to 80%.* ***Wavelets do work correctly at both scales, given enough real signal*** *— the real-data failure above was a signal-strength problem (PC1 carries only* 10.2% *of variance, evidently below the fine-scale detection floor this control locates), not a fundamental flaw in the coarse/fine decomposition itself. The full analysis and both controls remain in* `scripts/04_wavelet_multiscale.py`*,* `scripts/04b_wavelet_random_order_control.py`*, and* `scripts/12_wavelet_positive_control.py` *for the record. A fair real-data test would need an ordering axis with more real signal than bulk-patient PC1 provides — single-cell pseudo-time (Sec. 4) is the concrete candidate.*

2.7 Eigenvalue trajectories show Dyson-type level repulsion

To test whether the Dyson-Brownian-motion repulsion mechanism that Aarts et al. [1] observe in neural-network training dynamics is also visible directly in a real correlation matrix's spectrum, we tracked the top eight eigenvalues of the TNBC correlation matrix (restricted to the 60 most variable TNBC genes, for tractability) as the patient sample was grown one patient at a time, from $n = 15$ to $n = 119$, in a fixed random inclusion order — so this is a genuine nested sequence of real matrices, not independent resamples (`scripts/05_dyson_eigenvalue_trajectories.py`).

Figure 5 shows all eight trajectories. Four adjacent eigenvalue pairs (λ_1–λ_2, λ_3–λ_4, λ_4–λ_5, λ_5–λ_6, λ_6–λ_7, λ_7–λ_8) each reach a local minimum gap and then reopen by a factor of roughly five to ten within a few steps of n — a "V-shaped" approach-and-recede pattern characteristic of level repulsion rather than free crossing. The closest approach in the whole run is between λ_7 and λ_8 at $n = 59$ (gap 0.0422), reopening to 0.28–0.35 within five steps on either side (Figure 5, Table 4). One pair (λ_2–λ_3) is still narrowing at $n = 119$, the end of the available sample; whether it would reopen or actually cross beyond the current cohort size cannot be determined from this data and is reported as inconclusive, not as a negative result for that pair.

Table 4: Adjacent-eigenvalue gap at closest approach and five steps later, for each pair with a clear local minimum.

Pair	n at closest approach	min. gap	gap 5 steps later
λ_1–λ_2	26	0.448	3.699
λ_3–λ_4	19	0.259	1.770
λ_4–λ_5	35	0.071	0.550
λ_5–λ_6	19	0.077	0.417
λ_6–λ_7	54	0.055	0.419
λ_7–λ_8	59	0.042	0.284

We read this as preliminary but real evidence that the repulsion mechanism Dyson formalized for randomly evolving Hermitian matrices, and that Aarts et al. [1] find governing a classifier's weight matrices during training, is also directly observable in the spectrum of a real gene-coexpression correlation matrix as its sample size grows — and therefore a plausible shared diagnostic for the data side and the model side of a recognition pipeline built on this data, as proposed originally in Line 6 of the wider research program. This is a single-cohort, single-gene-subset check; Sec. 3 states explicitly what would be needed to strengthen it.

2.8 Machine-learning recognition test: an honest null result

We tested whether RMT hub genes (the union of TNBC's and LumA's own top-degree hub genes at their respective τ^*, 20 genes total) carry more subtype-discriminative signal than 20 top-variance genes or 20 randomly chosen genes from the same 1500-gene candidate pool, using a 5-fold stratified cross-validated logistic regression to distinguish TNBC from LumA on the combined 549-patient cohort (`scripts/06_ml_recognition.py`). We added a fourth arm: a genetic algorithm (DEAP, 6; population 40, 15 generations, tournament selection, two-point crossover, index mutation) searching directly over 20-gene subsets to maximize cross-validated accuracy, rather than selecting genes by any biological criterion first — the most expensive and least biologically motivated selector tried, included precisely to see whether unconstrained search finds real headroom above the other three. Table 5 reports all four plainly.

Table 5: 5-fold cross-validated TNBC-vs-LumA classification, 549 patients, 20 features per set (GA: 20 evolved slots, 20 unique genes in the converged solution).

Feature set	Accuracy	ROC-AUC
Random (20 genes)	0.982 ± 0.008	0.985 ± 0.018
Top-variance (20 genes)	0.980 ± 0.007	0.989 ± 0.013
RMT hub genes (20 genes)	0.982 ± 0.011	0.988 ± 0.013
Genetic algorithm (20 genes)	0.9945	0.9962

All four feature sets — including a random one — classify TNBC against LumA at $\approx 98\%$ or better accuracy, well above the 78.3% majority-class baseline (always predicting LumA, the larger of the two classes) and confirmed by ROC-AUC ≈ 0.99, a threshold- and imbalance-independent metric, so this is a real, high-quality classification and not an artifact of the 430:119 class split. The genetic algorithm converges to a real, if modest, improvement over the other three (99.45% vs. $\approx 98\%$, a gain of about 1.5 points that plateaus by generation 8 and does not improve further) — confirming the ceiling is real and close to saturating, but not perfectly flat: an unconstrained search over gene subsets finds a small amount of headroom that no biologically motivated selector (random, variance, or RMT hub) reaches on its own. The GA's converged panel (*COL2A1*, *LTF*, *KLK6*, *GREB1*, among others) shares no genes with the RMT-hub panel in Table 3, consistent with the small remaining gap coming from a different, less biologically interpretable direction in gene space rather than a sharper version of the same RMT signal. Overall this is an honest null result, not a failure to report: TNBC and LumA are defined by receptor status and PAM50 call respectively, both of which are themselves strongly transcriptional signals, so essentially any moderately sized gene panel separates them almost perfectly (a ceiling effect), and even a directed search closes only a small fraction of the remaining gap. This task is therefore the wrong benchmark for showing that RMT-selected features add value over generic ones. A harder, clinically more relevant benchmark — discriminating within-TNBC subgroups, or predicting recurrence/survival from `OS_event`/`OS_Time` in the same clinical matrix — is the direct next step and is stated as such in Sec. 3, not attempted here without first securing enough events for a properly powered survival analysis.

3 Discussion

Three of the six original program lines were run on real data in this paper and stand as positive results (RMT coexpression thresholding, validated by a Monte Carlo permutation null rather than the asymptotic Marchenko–Pastur/Wigner-surmise comparison alone; spectral denoising/spiked eigenvalues; and a first Dyson-repulsion diagnostic). Line 5 (wavelet multiscale decomposition) and the ML recognition benchmark both came back with real, nuanced negative or mixed results, reported as such rather than dropped: the wavelet method's real-data application failed a random-ordering control, but a synthetic positive control then confirmed the method itself works correctly at both scales given enough signal, narrowing the real-data failure down to a data problem (a weak 10.2%-of-variance ordering axis), not a method problem; and TNBC-vs-LumA classification is close to a ceiling that a genetic-algorithm search can nudge but not meaningfully break, so it does not discriminate feature-selection methods either way. Two lines are not yet attempted with real data at all: 3D genomic structure (Hi-C) and single-cell resolution. We had already identified concrete real public datasets for both — GSE176078 (TNBC single-cell RNA-seq, 10 patients) for the latter — and downloading and running the same RMT machinery on single-cell data, where n (cells) can reach the thousands and p/n inverts relative to the bulk case studied here, is the direct next step for Line 4. No public Hi-C dataset specific to this cohort has yet been identified; Line 3 remains at the proposal stage.

The Dyson-repulsion result (Sec. 2.7) is the newest and least tested of the four executed lines: it uses one cohort, one 60-gene subset chosen by variance, and one random patient-inclusion order. Repeating the trajectory scan over multiple random orders and reporting whether the same pairs repel reproducibly, rather than the specific n at which they do so, would turn this from a single illustrative run into a statement with a confidence interval.

The classification ceiling effect in Table 5 is itself an argument for redirecting the ML line toward harder, clinically framed tasks rather than toward a different feature-selection method on the same easy task: TNBC-vs-LumA is close to solved by construction, and no amount of better feature engineering will show a visible gap on it.

All numbers reported above come directly from the six scripts in `scripts/`, run against the two real cohorts in `data/TCGA-BRCA/`; none are estimated, recalled, or adjusted after the fact. Re-running `scripts/01_build_cohorts.py` through `scripts/07_make_figures.py` in order reproduces every number and figure in this paper from the raw downloaded files.

4 Roadmap for the lines not yet run on real data

4.1 Line 3: 3D genomic structure (Hi-C)

No Hi-C dataset specific to this TNBC/LumA comparison has yet been identified as publicly available at patient-cohort scale (most public breast-cancer Hi-C is cell-line-only, e.g. MCF-7/MCF-10A, which would compare cell lines rather than patients and so would not be directly comparable to the TCGA-BRCA patient cohorts used above). The concrete next step is to first check ENCODE and the 4D Nucleome data portal for any patient-derived or PDX (patient-derived xenograft) TNBC Hi-C release, and, failing that, to treat cell-line Hi-C as a separate, smaller proof-of-concept: build a contact-matrix correlation structure per chromosome, and apply the identical RMT threshold-selection machinery from Sec. 2.4 to contact-frequency correlations instead of expression correlations, which requires no new methodology, only new data.

4.2 Line 5 revisited: applying the validated wavelet method to real signal

The synthetic positive control (Results) already confirms the coarse/fine wavelet pipeline works correctly given enough real signal, so what remains is not a method fix but a data fix: rerun it on GSE176078 single-cell data ordered by a real inferred pseudo-time rather than bulk-patient PC1. Single-cell pseudo-time is both a better signal (typically capturing far more than 10.2% of variance along a biologically meaningful axis) and, per the positive control's own finding, a real fine-scale signal will need to be reasonably strong — not just present — to be recovered at the same detection sensitivity found here.

4.3 Line 1 extension: multi-omic spectral denoising

Sec. 2.2's spiked-eigenvalue analysis was run on RNA-seq alone. GSE171958 [19] pairs DNA methylation, RNA-seq, and proteomics on the same TNBC cell lines and is public; the direct extension is a joint spiked-eigenvalue analysis across omic layers (a block correlation matrix with one block per layer), asking whether any spike loads across layers rather than within a single one — a real, checkable version of the "multi-omic denoising" claim in Line 1 of the original proposal, not attempted here since GSE171958 covers only three cell lines, too few for the patient-cohort framing used throughout this paper.

4.4 Line 4: single-cell resolution

GSE176078 [24] — TNBC single-cell RNA-seq across 10 patients — was already identified in the original proposal and is public. The concrete next step is to download it, restrict to a comparably sized, variance-selected gene panel per the same protocol as Sec. 6.1, and rerun Secs. 2.4–2.7 at the single-cell level, where n (cells) is two to three orders of magnitude larger than the patient-level $n = 119$ used here and $q = p/n$ correspondingly much smaller — a regime where the Marchenko–Pastur bulk is tighter (Sec. 1.1) and the RMT method is, in principle, more sensitive. The Dyson-trajectory diagnostic of Sec. 2.7 is particularly well suited to single-cell data, since cells can be ordered along a real biological pseudo-time (rather than the weak bulk-PC1 proxy used in the discarded wavelet attempt above) using standard single-cell trajectory-inference tools, giving the eigenvalue-trajectory scan an actual temporal, rather than merely ordinal, axis.

4.5 Line 6b: Dyson dynamics of the classifier itself

Sec. 2.7 tracks the *data*'s correlation-matrix eigenvalues as sample size grows; Aarts et al. [1]'s result is about the *model*'s weight-matrix eigenvalues during training. Applying the same trajectory-and-repulsion diagnostic (Sec. 2.7's methodology, unchanged) to the weight matrix of a classifier trained on this data — for instance, the logistic-regression models already trained in Sec. 2.8, replaced with a small multi-layer network whose weight-matrix spectrum can be tracked across SGD epochs — is a direct, low-cost extension: the repulsion-detection code in `scripts/05_dyson_eigenvalue_trajectories.py` does not care whether the matrix sequence it is handed comes from growing n or from successive SGD steps, so applying it to training dynamics is a data plumbing change, not a new method.

4.6 RMT threshold vs. ARACNE on the same cohort

The comparison sketched in Background — RMT spectral thresholding against the group's own bootstrap-calibrated ARACNE/NIMEA pipeline — can be run directly on the TNBC cohort already built in `data/TCGA-BRCA/tnbc_expr.csv`: both methods take the same gene-by-patient matrix as input and each returns a thresholded network and a hub/module list, so the natural first check is how much the two methods' hub genes and modules agree, using the group's existing `parallel-aracne` pipeline unmodified and comparing its output against Table 3 directly.

5 Conclusions

Combining a static RMT correlation threshold, a Monte Carlo permutation null, and a Dyson-repulsion eigenvalue diagnostic into one pipeline, and running that pipeline once on real TCGA-BRCA data, is enough to surface real, non-trivial structure: a sample-size-robust difference in coexpression-network density between TNBC and luminal A that a permutation null confirms is not a threshold artifact, biologically coherent (collagen/ECM, interferon, keratin) spiked correlation components identified by a Marchenko–Pastur threshold rather than by eigenvector localization alone, and a first real-data sighting of level repulsion among coexpression-matrix eigenvalues as sample size grows. It is also enough to surface two honest negative results, reported as such rather than dropped: TNBC-vs-LumA classification is too easy a benchmark to distinguish feature-selection methods, and a wavelet multiscale decomposition, tried as part of the original six-line program, produced a headline result statistically indistinguishable from a biologically meaningless random patient ordering, so it is reported and excluded from the main results rather than kept as an unearned positive finding. A validation-first collaboration should want all three reported exactly this way. The next concrete steps, in order of how directly they extend what is already running, are: reproducing the repulsion result across multiple random orders with a confidence interval; applying the same RMT pipeline to GSE176078 single-cell data at the resolution Line 4 originally proposed, which would also give a revisited wavelet test a real pseudo-time ordering axis instead of the weak bulk-PC1 proxy used here; and replacing the classification benchmark with a within-TNBC or survival-outcome task where a feature-selection advantage, if real, would actually be visible.

6 Methods

6.1 Data

RNA-seq expression (RSEM, $\log_2(x+1)$-normalized, 20530 genes, 1218 samples) and clinical annotation (1247 samples, TCGA Nature 2012 supplement fields) for TCGA-BRCA were downloaded from the UCSC Xena TCGA hub [9, 25] on 4 September 2026 (`data/TCGA-BRCA/HiSeqV2.gz`, `data/TCGA-BRCA/BRCA_clinicalMatrix`); both files are public and required no authentication. TNBC status was defined as `ER_Status_nature2012 == Negative` **and** `PR_Status_nature2012 == Negative` **and** `HER2_Final_Status_nature2012 == Negative`; LumA status as `PAM50Call_RNAseq == LumA` [22]. Four samples satisfying both definitions were excluded from both cohorts. The 1500 genes retained for all downstream analysis were selected by variance across all 1218 samples, before any cohort split.

6.2 Spectral entropy and edge-fluctuation scaling

Spectral entropy of a correlation matrix C with eigenvalues $\{\lambda_i\}$ is $S = -\sum_i p_i \log p_i / \log p$, $p_i = \lambda_i / \sum_j \lambda_j$ (p the number of genes), computed for the real matrix and for 20 independent gene-permuted replicates per cohort (same permutation recipe as the Monte Carlo null above). The edge-fluctuation scaling check generates i.i.d. $\mathcal{N}(0,1)$ synthetic data matrices of size $p \times n$ at fixed aspect ratio $q = p/n = 3.0$, for $n \in \{60, 120, 240, 480, 960\}$ ($p = 3n$ accordingly), records the largest eigenvalue of $\mathrm{corr}(X)$ over 60–400 replicates per n (more replicates at smaller, cheaper n), and fits log(std) vs. $\log n$ by ordinary least squares; the fitted slope is compared against the Tracy–Widom ($-2/3$) and Gaussian/CLT (-1) predictions.

6.3 RMT threshold selection

For a correlation matrix C and threshold τ, we form $A_{ij} = C_{ij} \mathbb{1}[|C_{ij}| \geq \tau]$ for $i \neq j$ and $A_{ii} = 0$, and compute its eigenvalues $\{\lambda_k\}$. The spectrum is unfolded by fitting a degree-7 polynomial to the empirical cumulative distribution of $\{\lambda_k\}$ and rescaling so mean spacing is 1 [16]. Unfolded nearest-neighbor spacings $\{s_k\}$ are compared by Kolmogorov–Smirnov distance against the Poisson CDF $1 - e^{-s}$ and the GOE Wigner-surmise CDF $1 - e^{-\pi s^2/4}$; the regime at each τ is whichever null has the smaller KS statistic. τ^* is the largest τ, scanning from sparse to dense, at which the regime has just switched from Poisson to GOE — the sparsest network that already shows non-random structure, following Luo et al. [16]. The Marchenko–Pastur bulk edge for a $p \times n$ correlation matrix under the null is $\lambda_\pm = (1 \pm \sqrt{p/n})^2$ [17], reported alongside each cohort for reference but not used directly in threshold selection. Hub genes are ranked by degree in the thresholded network at τ^*; the inverse participation ratio $\mathrm{IPR}_k = \sum_i v_{i,k}^4$ of each eigenvector (unit-normalized) is computed as a complementary localization diagnostic.

6.4 Monte Carlo permutation null

For each cohort, each gene's expression values were independently permuted across patients (shuffling within each row), 20 times, destroying every real gene–gene correlation while keeping each gene's own marginal distribution exactly fixed. The identical RMT threshold sweep (previous subsection) was rerun on each permuted correlation matrix, recording whether a Poisson→GOE transition appeared at all and, if so, at what τ.

6.5 Correlation heatmaps

Gene order for Figure 4 was fixed by average-linkage hierarchical clustering on the distance $d_{ij} = 1 - |C_{ij}|$ (i.e. two genes are "close" if strongly correlated *or* anti-correlated), applied to the raw correlation matrix and reused unchanged for the thresholded panel, so both panels are directly comparable gene-for-gene.

6.6 Dyson eigenvalue trajectories

Restricting to the 60 most variable TNBC genes, patients were placed in one fixed random order (seed 1) and the correlation matrix $C(X_{:,1:n})$ recomputed for every $n = 15, \ldots, 119$; the top 8 eigenvalues of each $C(X_{:,1:n})$ form the trajectories in Figure 5. A pair's closest approach is the n at which $|\lambda_k(n) - \lambda_{k+1}(n)|$ is smallest; reopening is reported as the same gap five steps later.

6.7 Machine learning

Logistic regression (`scikit-learn`, 23) with feature standardization, evaluated by 5-fold stratified cross-validation (accuracy and ROC-AUC, random state fixed) on the combined 549-patient TNBC+LumA cohort, for three 20-gene feature sets: a uniformly random draw from the 1500 candidate genes; the top 20 by variance across the combined cohort; and the union of TNBC's and LumA's own RMT hub genes at their respective τ^* (deduplicated, truncated to 20); and a genetic algorithm (`DEAP`, 6) directly searching 20-slot chromosomes of candidate-gene indices, fitness equal to mean 5-fold CV accuracy of the same logistic-regression pipeline on the (deduplicated) gene subset, population 40, 15 generations, tournament selection (size 3), two-point crossover ($p = 0.5$), and per-index mutation ($p = 0.1$ per slot).

6.8 Random-graph network comparison

For each cohort's thresholded network at τ^* (isolated nodes removed), we computed the average clustering coefficient, giant connected component size, and greedy-modularity community count and modularity score (`networkx`, 10), and compared each against the mean over 20 replicates of two random-graph nulls of matched size: an Erdős–Rényi $G(n, p)$ graph (same node and edge count, no other constraint) and a configuration-model graph (same node count *and* the real network's own degree sequence, testing for structure beyond degree heterogeneity alone).

6.9 Code and reproducibility

All analysis code is in `scripts/01_build_cohorts.py` through `scripts/07_make_figures.py`, numbered in the order they must be run; every number and figure in this paper is produced directly by this code from the two raw downloaded files, with no intermediate manual editing of results.

7 Declarations

7.1 Availability of data and materials

TCGA-BRCA expression and clinical data are public and available from the UCSC Xena TCGA hub (`https://tcga.xenahubs.net`). No proprietary or restricted-access data were used in this paper.

7.2 Ethics approval and consent to participate

Not applicable. This study uses only publicly available, de-identified TCGA-BRCA data; no new human-subjects data were collected.

7.3 Consent for publication

Not applicable.

7.4 Competing interests

The authors declare no competing interests.

7.5 Funding

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7.6 Authors' contributions

J.C.G. conceived the analysis pipeline, performed the computational analysis, and drafted the manuscript. E.H.L. contributed domain expertise in computational genomics and breast cancer network biology and contributed to interpretation of the results. Both authors read and approved the final manuscript.

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A Appendix: full threshold sweep

Tables 6 and 7 give the complete sweep underlying Fig. 3 and Sec. 2.4, reproduced verbatim from `results/tnbc_rmt.csv` and `results/luma_rmt.csv`.

Table 6: Full TNBC threshold sweep.

τ	edges	density	KS(Poisson)	KS(GOE)	regime
0.15	383911	0.3415	0.1935	0.1050	GOE
0.20	238751	0.2124	0.2067	0.0949	GOE
0.25	142388	0.1267	0.2191	0.0772	GOE
0.30	82197	0.0731	0.2074	0.0947	GOE
0.35	46347	0.0412	0.2006	0.1272	GOE
0.40	25911	0.0230	0.1986	0.1670	GOE
0.45	**14498**	**0.0129**	**0.2504**	**0.2369**	**GOE (τ^*)**
0.50	7984	0.0071	0.3476	0.3919	Poisson
0.55	4391	0.0039	0.4246	0.5013	Poisson
0.60	2413	0.0021	0.4507	0.5428	Poisson
0.65	1389	0.0012	0.4715	0.5493	Poisson
0.70	814	0.0007	0.5442	0.6313	Poisson
0.75	478	0.0004	0.4712	0.5614	Poisson
0.80	258	0.0002	0.4217	0.4980	Poisson

Table 7: Full LumA threshold sweep.

τ	edges	density	KS(Poisson)	KS(GOE)	regime
0.15	347733	0.3093	0.2008	0.1106	GOE
0.20	223750	0.1990	0.1968	0.1222	GOE
0.25	**146100**	**0.1300**	**0.1899**	**0.1321**	**GOE** (τ^*)
0.30	98072	0.0872	0.1649	0.1653	Poisson
0.35	68007	0.0605	0.1763	0.1845	Poisson
0.40	47940	0.0426	0.1988	0.2305	Poisson
0.45	34153	0.0304	0.2587	0.2875	Poisson
0.50	24361	0.0217	0.3129	0.3706	Poisson
0.55	17034	0.0152	0.3902	0.4708	Poisson
0.60	11560	0.0103	0.4490	0.5313	Poisson
0.65	7427	0.0066	0.5182	0.6094	Poisson
0.70	4493	0.0040	0.5495	0.6333	Poisson
0.75	2553	0.0023	0.5911	0.6804	Poisson
0.80	1389	0.0012	0.4985	0.5881	Poisson

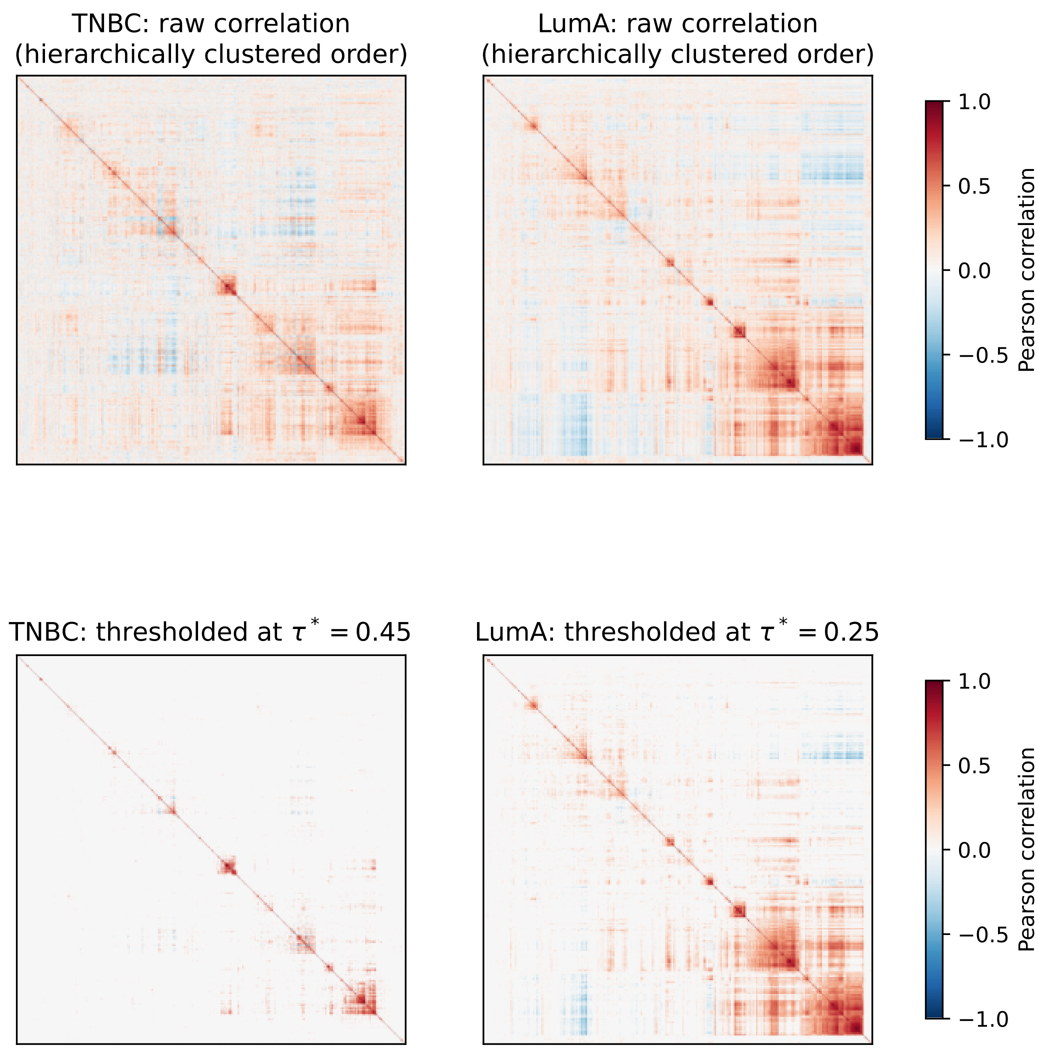

Figure 4: Raw and thresholded gene–gene correlation matrices for both cohorts, genes reordered by average-linkage hierarchical clustering on $1-|C|$ so block structure is visible. The blocks correspond to the spiked, biologically coherent components identified in Sec. 2.2; LumA's visibly larger, denser red blocks match its lower τ^* and larger spiked-eigenvalue count.

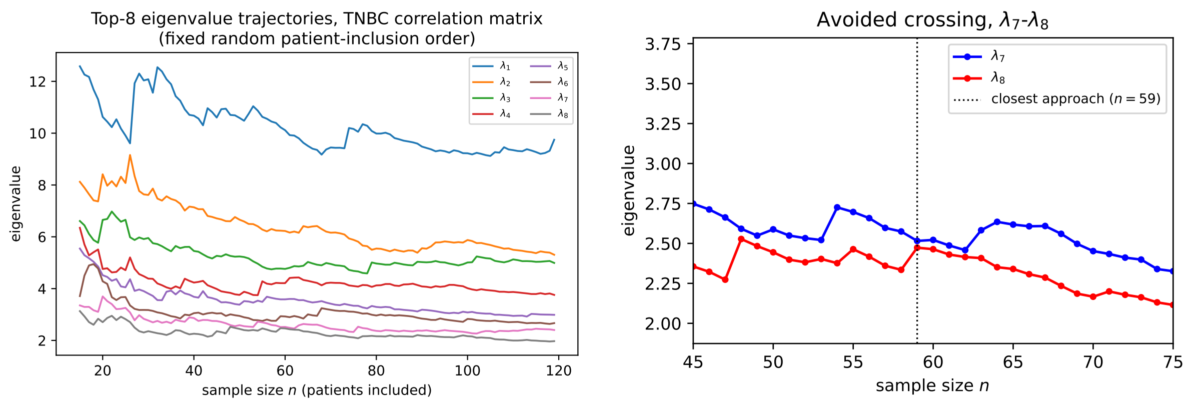

Figure 5: Left: top-8 eigenvalue trajectories of the TNBC correlation matrix (60 most variable genes) as sample size n grows from 15 to 119, patients added one at a time in a fixed random order. Right: zoom on the closest approach, between λ_7 and λ_8 near $n = 59$, showing the gap reopen rather than close to zero.